

University College London
Department of Economics
Main Summer Assessment Period 2025

ECON0016
Macroeconomic Theory and Policy

Student Name:

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Student Number:

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Time allowance: You have **THREE (3) hours** to complete this examination.

If you have been granted SoRA extra time and/or rest breaks, your individual examination duration will be extended pro-rata.

Calculators (UCL approved models only) **are** permitted in this examination.

Number of Questions Answered Policy: In cases where a student answers more questions than requested by the examination rubric, the policy of the Department of Economics is that the student's first set of answers up to the required number will be the ones that count (not the best answers). All remaining answers will be ignored.

Recording of answers:

- Use the separate paper provided to do all the necessary rough work, calculations or drafting of answers. If you need more paper for rough work, alert the invigilators by raising your hand.
- Enter your final answers, answer steps, and explanation/interpretation (as required) in the answer-boxes provided within this question-and-answer booklet.
- **Only results, workings, diagrams, and writing provided within the appropriate boxed answer-spaces will be marked.** Comments, notes or anything else outside of the provided boxed answer-spaces will be ignored.
- **Do not** remove any pages from this answer booklet!
- If you inadvertently damage your answer booklet during the exam, alert the invigilators and request a new answer booklet. You will **not** get additional time!

Submission of answers:

- Ensure that you have provided in the spaces above your **correct student number** (see your student card) and your **correct name**, in block capitals. Failure to provide this information may mean that any marks awarded cannot be allocated to you correctly.
- Only submit your answer booklet! Submitting a damaged or incomplete answer booklet may result in a mark of 0.

The remainder of this page has been left blank intentionally. Do not use the space on this page to record your answers to questions in this exam.

Exam Instructions and Questions

You must answer ALL questions from those presented in the remainder of this booklet. Specifically, you must answer SIX questions from Part A, FOUR questions from Part B, THREE questions from Part C, SEVEN questions from Part D and SIX questions from Part E.

Marks for each subquestion are specified in the instructions for that question.

The remainder of this page has been left blank intentionally. Do not use the space on this page to record your answers to questions in this exam.

Part A: True/False/Uncertain Questions (24 points)

Answer ALL questions from Part A by specifying whether the statement is TRUE/FALSE/-UNCERTAIN. For each question, write a concise explanation and include equations and graph(s) if needed (you can use formulas and diagrams, but you MUST write your argument in words too). Make sure to clearly mention on which theory/model your argument is based: maximum marks will be given for a correct explanation that explicitly reference the course material. Each question is worth 4 points. Record your answers in the space below the questions. If you need more space, use the ADDITIONAL SPACE at the end of the booklet, indicating clearly to which question(s) your answer pertains.

A1. In the 3-Equations Model, inflation expectations can determine only the intercept of the Phillips Curve, as they matter for the relationship between current inflation and output.

Record your answer to Question A1 in the space below. Do NOT use this space to record answers to any questions other than A1.

PART A cont'd: Answer ALL questions from Part A

A2. Depending on the conduct of the monetary policy, the multiplier of fiscal interventions can be zero.

Record your answer to Question A2 in the space below. Do NOT use this space to record answers to any questions other than A2.

PART A cont'd: Answer ALL questions from Part A

A3. When the government announces a future fiscal deficit reduction, expected future output falls, but current output may increase.

Record your answer to Question A3 in the space below. Do NOT use this space to record answers to any questions other than A3.

PART A cont'd: Answer ALL questions from Part A

A4. Quantitative easing means that Central Bank issues more long-term assets, so they can decrease the long-term interest rate and reinforce the confidence of the public in the state of the economy.

Record your answer to Question A4 in the space below. Do NOT use this space to record answers to any questions other than A4.

PART A cont'd: Answer ALL questions from Part A

A5. An economy can stabilize its debt-to-GDP ratio only by running a primary fiscal surplus.

Record your answer to Question A5 in the space below. Do NOT use this space to record answers to any questions other than A5.

PART A cont'd: Answer ALL questions from Part A

A6. Under the permanent income hypothesis, individuals who are unable to save will respond more strongly to news that taxes will increase in the future compared to other households.

Record your answer to Question A6 in the space below. Do NOT use this space to record answers to any questions other than A6.

Part B: Long Exercise (26 points)

Answer **ALL** questions of the following Part B exercise. Each question is worth 6.5 points. Write your answers in words and include equations and graph(s) if needed or specified by the question. Record your answer to each question in the space below the question. If you need more space, use the **ADDITIONAL SPACE** at the end of the booklet, indicating clearly to which question(s) your answer pertains.

To start, recall the expressions for the wage setting (WS) and price setting (PS) curves:

$$WS : \quad \frac{W}{P} = 1 - \alpha u + z$$
$$PS : \quad \frac{W}{P} = \frac{\lambda}{(1 + \tau)(1 + \mu)}$$

Consider an economy at the medium run equilibrium (MRE). Suppose international events cause an increase in the price of natural gas. Simultaneously, this also causes changes in other commodity prices.

B1. Is this a demand- or a supply-side shock? Does it affect the WS or the PS curve? Can you briefly name and comment on at least 3 other shocks that can affect the WS and/or the PS curves?

Record your answer to Question B1 in the space below. Do NOT use this space to record answers to any questions other than B1.

Additional answer-space for Question B1.

PART B cont'd: Answer ALL questions from Part B

B2. Starting from the MRE, show the response of the economy to a permanent increase in natural gas and other commodity prices using the 3-Equations model, and carefully explain the path of output, inflation and the real interest rate. Please be explicit with what happens in each period, for at least 2 periods after the shock. All three diagrams studied in class should be present for full marks (you can re-use the WS-PS one, if you have already drawn it in B1).

Record your answer to Question B2 in the space below. Do NOT use this space to record answers to any questions other than B2.

Additional answer-space for Question B2.

PART B cont'd: Answer ALL questions from Part B

B3. Assume that the economy in question is an oil producer. Therefore, the shock to gas prices improves business expectations permanently. Analyze the shock through the lenses of our model by focusing on this effect and neglecting the one suggested in B2. Which curve(s) shift? What is the new MRE? Note: you do not need to re-draw all the diagrams.

Record your answer to Question B3 in the space below. Do NOT use this space to record answers to any questions other than B3.

PART B cont'd: Answer ALL questions from Part B

B4. As a reaction to your analysis in B2, the government approves a reform which decreases the frequency of bargaining rounds for workers. This reform implies that wages are stickier than prices now. Does it affect any curve in the 3-Equations model? Explain only the intuition behind this effect. Note: you do not need to re-draw all the diagrams.

Record your answer to Question B4 in the space below. Do NOT use this space to record answers to any questions other than B4.

Part C: True/False/Uncertain Questions (15 points)

Answer ALL questions from Part C by specifying whether the statement is TRUE/FALSE/-UNCERTAIN. For each question, write a concise explanation and include equations and graph(s) if needed (you can use formulas and diagrams, but you MUST write your argument in words too). Make sure to clearly mention on which theory/model your argument is based: maximum marks will be given for a correct explanation that explicitly reference the course material. Each question is worth 5 points. Record your answers in the space below the questions. If you need more space, use the ADDITIONAL SPACE at the end of the booklet, indicating clearly to which question(s) your answer pertains.

C1. According to the return differential hypothesis, China pays a higher rate of return on its international liabilities than what it receives on its international assets.

Record your answer to Question C1 in the space below. Do NOT use this space to record answers to any questions other than C1.

PART C cont'd: Answer ALL questions from Part C

C2. A country starts 2022 as a net creditor. The interest rate on its net asset position is 10 percent. That year, it runs a current account deficit. It follows that the trade balance in 2022 was also negative.

Record your answer to Question C2 in the space below. Do NOT use this space to record answers to any questions other than C2.

PART C cont'd: Answer ALL questions from Part C

C3. Arezki, Ramey, and Sheng (Quarterly Journal of Economics, 2017) have studied the economic impact of giant oil discoveries. They have shown that the current account improves for 5 years and then experiences a reversal with a trough in year 8. This is in line with the intertemporal theory of the current account in an economy with production.

Record your answer to Question C3 in the space below. Do NOT use this space to record answers to any questions other than C3.

PART D: Long Exercise (15 points)

Answer **ALL** questions of the following Part D exercise. Each question is worth $15/7 \approx 2.14$ points. Write your answers in words and include equations if needed or specified by the question. Record your answer to each question in the space below the question. If you need more space, use the **ADDITIONAL SPACE** at the end of the booklet, indicating clearly to which question(s) your answer pertains.

This is about international comparison of standards of living. The following table displays (fictional) quantities of final goods produced and prices in the United Kingdom and Morocco in 2025.

Good	United Kingdom		Morocco	
	Quantity	Price	Quantity	Price
Tradable	10	5	0.4	50
Nontradable	20	15	0.8	100

U.K. prices are expressed in pounds and Morocco prices in dirhams. The typical basket of goods in both countries includes 1 unit of tradable goods and 2 units of nontradable goods. Tradable goods are freely traded internationally. For this type of good, Purchasing Power Parity (PPP) holds.

D1. Calculate the market exchange rate, expressed as the pound price of one dirham.

Record your answer to Question D1 in the space below. Do NOT use this space to record answers to any questions other than D1.

PART D cont'd: Answer ALL questions from Part D

D2. Calculate the price levels in the United Kingdom and Morocco, expressed in pounds and dirhams, respectively.

Record your answer to Question D2 in the space below. Do NOT use this space to record answers to any questions other than D2.

PART D cont'd: Answer ALL questions from Part D

D3. What is the pound/dirham real exchange rate? In which country is it cheaper to live?

Record your answer to Question D3 in the space below. Do NOT use this space to record answers to any questions other than D3.

PART D cont'd: Answer ALL questions from Part D

D4. Calculate GDP in pounds in the United Kingdom and Morocco at market prices. According to this measure, how big is Morocco relative to the United Kingdom?

Record your answer to Question D4 in the space below. Do NOT use this space to record answers to any questions other than D4.

PART D cont'd: Answer ALL questions from Part D

D5. Calculate the PPP exchange rate.

Record your answer to Question D5 in the space below. Do NOT use this space to record answers to any questions other than D5.

PART D cont'd: Answer ALL questions from Part D

D6. Calculate both GDPs in pounds at PPP prices.

Record your answer to Question D6 in the space below. Do NOT use this space to record answers to any questions other than D6.

PART D cont'd: Answer ALL questions from Part D

D7. Comment on the numbers you obtained for the two measures of GDP.

Record your answer to Question D7 in the space below. Do NOT use this space to record answers to any questions other than D7.

PART E: Long Exercise (20 points)

Answer ALL questions of the following Part E exercise. Each sub-question is worth 2.5 points except E4 which is worth 7.5 points. Write your answers in words and include equations if needed or specified by the question. Record your answer to each question in the space below the question. If you need more space, use the **ADDITIONAL SPACE** at the end of the booklet, indicating clearly to which question(s) your answer pertains.

Consider a two-period, small open endowment economy populated by a large number (normalised to 1) of households with preferences described by the utility function

$$\log C_1 + \beta \log C_2,$$

where C_1 and C_2 denote consumption in periods 1 and 2, and β is the subjective discount factor. Households receive endowments Q_1 in period 1 and Q_2 in period 2, and can borrow or lend in international financial markets at the interest rate r^* . The government imposes taxes $T_1 = T^L + \tau_1 C_1$ in period 1 (T^L are lump sum taxes) and $T_2 = \tau_2 C_2$ in period 2 and consumes G_1 units of goods in period 1 and G_2 units in period 2. Finally, households and the government start period 1 with no assets or debts carried over from the past.

E1. Derive the intertemporal budget constraint of the household, the intertemporal budget constraint of the government, and the intertemporal resource constraint of the economy as a whole.

Record your answer to Question E1 in the space below. Do NOT use this space to record answers to any questions other than E1.

Additional answer-space for Question E1.

PART E cont'd: Answer ALL questions from Part E

E2. Derive the optimality condition that results from choosing C_1 and C_2 to maximize the household's utility function subject to the household's intertemporal budget constraint.

Record your answer to Question E2 in the space below. Do NOT use this space to record answers to any questions other than E2.

PART E cont'd: Answer ALL questions from Part E

E3. Define an equilibrium of this small open economy.

Record your answer to Question E3 in the space below. Do NOT use this space to record answers to any questions other than E3.

PART E cont'd: Answer ALL questions from Part E

E4. Assume τ_1 and τ_2 are given. Find an expression for the equilibrium levels of consumption C_t and the trade balance TB_t in periods 1 and 2, the equilibrium level of lump-sum taxes T^L and the equilibrium primary fiscal deficit in period 1 DEF_1 .

Record your answer to Question E4 in the space below. Do NOT use this space to record answers to any questions other than E4.

PART E cont'd: Answer ALL questions from Part E

When we assume $\beta = 1/1.1$, $Q_1 = Q_2 = 10$, $G_1 = G_2 = 2$, $\tau_1 = \tau_2 = .2$ and $r^* = .1$, we find the equilibrium values displayed in Table 1 (column E.4).

Table 1

	Question E4	Question E5	Question E6
τ_1	0.20	0.10	0.10
τ_2	0.20	0.33	0.20
T^L	0.76	0.76	1.60
C_1	8.00	8.73	8.33
C_2	8.00	7.20	7.64
TB_1	0.00	-0.73	-0.33
TB_2	0.00	0.80	0.36
DEF_1	-0.36	0.36	-0.43

E5. Continue to assume that $G_1 = G_2 = 2$. Suppose that the government implements a tax cut in period 1 consisting in lowering the consumption tax rate from 20 to 10 percent. Suppose further that lump-sum taxes, T^L , are kept at the level found in the previous item, and τ_2 is adjusted to balance the government intertemporal budget constraint. Equilibrium values are displayed in Table 1 (column E5). Comment the difference between E4 and E5.

Record your answer to Question E5 in the space below. Do NOT use this space to record answers to any questions other than E5.

PART E cont'd: Answer ALL questions from Part E

E6. Assume now that the cut in consumption taxes in period 1 from 20 to 10 percent is financed with an appropriate change in lump-sum taxes T^L in the same period, while the consumption tax rate in period 2 is kept constant at its initial level of 20 percent. Equilibrium values are displayed in Table 1 (column E.6). Comment the difference between E5 and E6.

Record your answer to Question E6 in the space below. Do NOT use this space to record answers to any questions other than E6.

If you need extra space for a question, you may use the Additional Space on this page should you require it, indicating clearly to which question(s) your answers pertain.

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End of Exam Checklist:

Please ensure that:

- You have correctly written your student number and name on the cover sheet (page 1).
- Your answer-booklet is complete and not damaged.

The remainder of this page has been left blank intentionally. Do not use the space on this page to record your answers to questions in this exam.